

Transformation in ETL - Assignment Solution

Question 1

Data Transformation is the process of converting raw data into a clean, consistent, and usable format during the ETL process. It is important because it improves data quality, consistency, and accuracy for analysis and reporting.

Question 2

Four common data cleaning activities are: (1) Handling missing values, (2) Removing duplicate records, (3) Correcting inconsistent data formats, and (4) Identifying and treating outliers.

Question 3

Normalization scales data to a fixed range, usually 0 to 1. Standardization transforms data so that it has a mean of 0 and a standard deviation of 1. Normalization is useful when data ranges differ greatly, while standardization is preferred for many machine learning algorithms.

Question 4

Two techniques to handle missing Age values are: (1) Mean/Median Imputation – use when only a few values are missing. (2) Row Deletion – use when the number of missing values is very small and removing rows will not affect analysis.

Question 5

Standardized Gender Values:

[M, male, F, Female, MALE, f] → [Male, Male, Female, Female, Male, Female]

Question 6

One-Hot Encoding converts categorical values into binary columns.

Example:

Red = [1,0,0]

Blue = [0,1,0]

Green = [0,0,1]

Question 7

Data Integration combines data from multiple sources into a unified dataset. Data Mapping defines how fields from one source correspond to fields in another system during integration.

Question 8

Z-score Standardization is preferred when outliers exist because it uses the mean and standard deviation, preserving the influence of extreme values. Min-Max Scaling can be heavily distorted by outliers since it depends on minimum and maximum values.